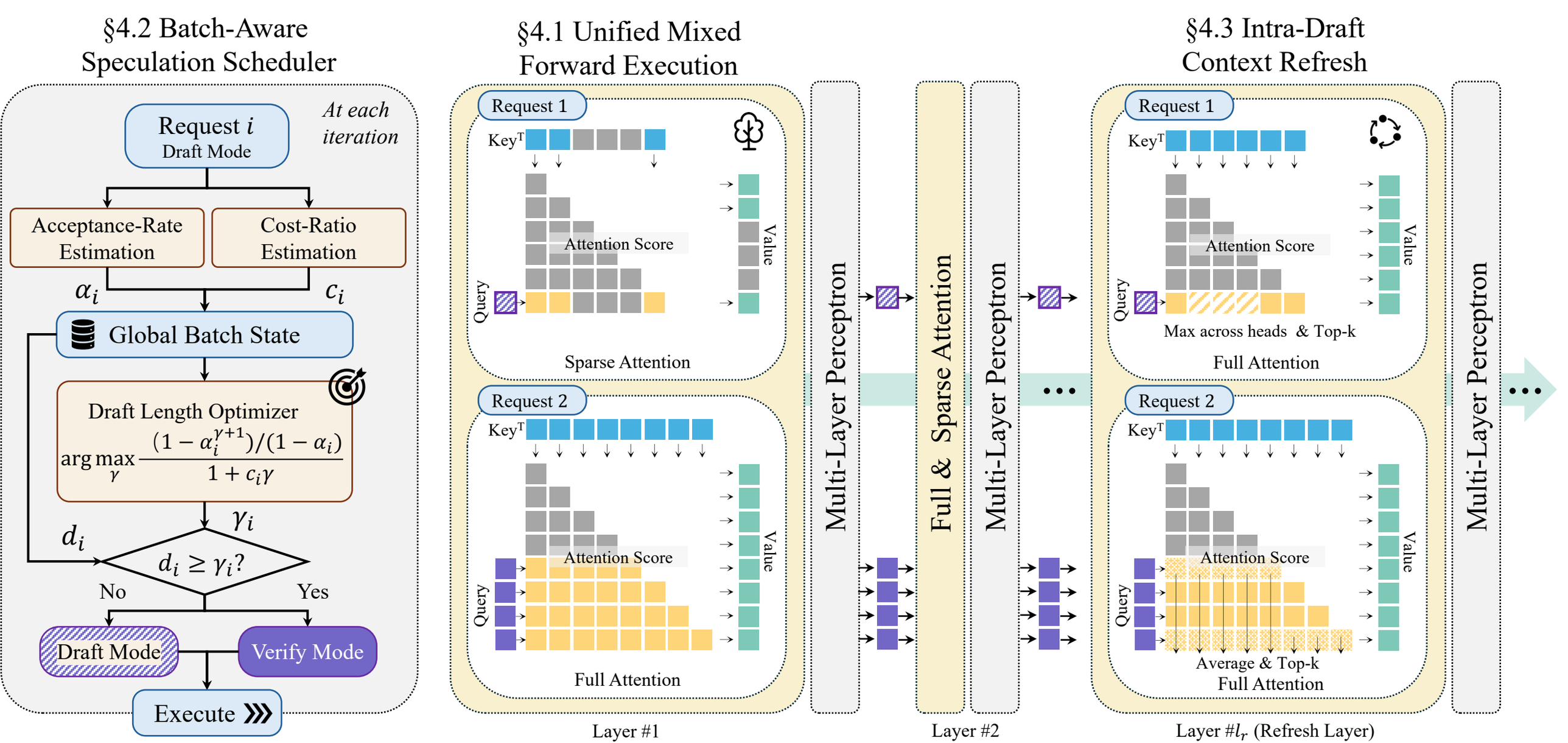

§4.2 Batch-Aware Speculation Scheduler
At each iteration
Request i
Draft Mode
Acceptance-Rate Estimation
Cost-Ratio Estimation
α_i
c_i
Global Batch State
Draft Length Optimizer
$\arg\max_{\gamma} \frac{(1-\alpha_i^{\gamma+1})/(1-\alpha_i)}{1+c_i\gamma}$
γ_i
d_i
$d_i \geq \gamma_i$?
No
Yes
Draft Mode
Verify Mode
Execute
§4.1 Unified Mixed Forward Execution
Request 1
Key^T
Query
Attention Score
Value
Sparse Attention
Request 2
Full Attention
Multi-Layer Perceptron
Layer #1
Full & Sparse Attention
Multi-Layer Perceptron
Layer #2
§4.3 Intra-Draft Context Refresh
Request 1
Max across heads & Top-k
Full Attention
Request 2
Average & Top-k
Full Attention
Multi-Layer Perceptron
Layer #l_r (Refresh Layer)